3.2.3

Wednesday, April 8, 2026 5:38 PM

Exercise 8 (Extended Euclidean Algorithm). Find integers $s,t \in \mathbb{Z}$ such that $gcd(a,b) = s \cdot a + t \cdot b$ holds for the following pairs:

- $(a,b) = (45,10)$
- $(a,b) = (13,11)$
- $(a,b) = (13,12)$

Exercise 9 (Towards Prime fields). Let $n \in \mathbb{N}$ be a natural number and p a prime number, such that $n < p$. What is the greatest common divisor $gcd(p,n)$?

Exercise 10. Find all numbers $k \in \mathbb{N}$ with $0 \leq k \leq 100$ such that $gcd(100,k) = 5$.

Exercise 11. Show that $gcd(n,m) = gcd(n+m,m)$ for all $n,m \in \mathbb{N}$.

8 (45,10) (13,11) (13,12)

k	r	s	t	q
0	45	1	0	
1	10	0	1	
2	5	1	-4	4
	0	2	9	2

gcd = 5
s = 1
t = -4

$5 = 1 \cdot 45 - 4 \cdot 10$

	13	1	0	
	11	0	1	
2	2	1	-1	1
	1	-5	6	5
	0	11	-13	2

gcd = 1
s = -5
t = 6

$1 = -5 \cdot 13 + 6 \cdot 11$

13	1	0	
12	0	1	
1	1	-1	1
0			12

gcd = 1
s = 1
t = -1

$1 = 1 \cdot 13 + -1 \cdot 12$

9 $gcd(p,n) = 1$

10 $100 = 2^2 \cdot 5^2$

2 50
2 25
5 5

$5 \cdot 3, 5 \cdot 7, 5 \cdot 9, 5 \cdot 11, 5 \cdot 13, 5 \cdot 17, 5 \cdot 19$
$5 \cdot 1$

11 $n = e \cdot i$ $\quad n + m = p \cdot e$
$m = e \cdot k$ $\quad m = q \cdot e$

$$m = e \cdot k$$

$$+ \qquad m = q \cdot e$$

$$n + m = e \cdot i + e \cdot k$$
$$= e \cdot (k + i)$$

$$-$$

$$n = (p - q) \cdot e$$

$$e = \gcd(n, m)$$